

Normalization of Database Tables

Database Tables and Normalization

- Table is basic building block in database design
- Table's structure is of great interest
- Two cases:
 - possible poor table structures in good database design
- Normalization can help recognize a poor table and convert to good tables with good structure

Database Tables and Normalization

- Normalization is process for re-assigning attributes to entities
 - Reduces data **redundancies**
 - Helps eliminate **data anomalies**
 - Produces **controlled** redundancies to link tables
 - Series steps called normal forms

Normalization and Database Design

- Normalization should be part of the design process
- Make sure the proposed entities meet the required normal form before the table structures are created
- Used to redesign or modify the existing table structures.
- E-R Diagram provides macro view

Data Normalization

- Primarily a tool to validate and improve a logical design so that it satisfies certain constraints that *avoid unnecessary duplication of data*
- The process of decomposing relations with anomalies to produce smaller, *well-structured* relations

Example – Figure 5.2b

EMPLOYEE2

<u>Emp_ID</u>	Name	Dept_Name	Salary	<u>Course_Title</u>	Date_Completed
100	Margaret Simpson	Marketing	48,000	SPSS	6/19/200X
100	Margaret Simpson	Marketing	48,000	Surveys	10/7/200X
140	Alan Beeton	Accounting	52,000	Tax Acc	12/8/200X
110	Chris Lucero	Info Systems	43,000	SPSS	1/12/200X
110	Chris Lucero	Info Systems	43,000	C++	4/22/200X
190	Lorenzo Davis	Finance	55,000		
150	Susan Martin	Marketing	42,000	SPSS	6/19/200X
150	Susan Martin	Marketing	42,000	Java	8/12/200X

Question – Is this a relation?

Answer – Yes: unique rows and no multivalued attributes

Question – What's the primary key?

Answer – Composite: Emp_ID, Course_Title

Well-Structured Relations

- A relation that contains minimal data redundancy and allows users to insert, delete, and update rows without causing data inconsistencies
- Goal is to avoid anomalies (next slide)
 - **Insertion Anomaly** – adding new rows forces user to create duplicate data
 - **Deletion Anomaly** – deleting rows may cause a loss of data that would be needed for other future rows
 - **Modification Anomaly** – changing data in a row forces changes to other rows because of duplication

General rule of thumb: a table should not pertain to more than one entity type

Anomalies in this Table

- **Insertion** – can't enter a new employee without having the employee take a class
- **Deletion** – if we remove employee 140, we lose information about the existence of other classes
- **Modification** – giving a salary increase to employee 100 forces us to update multiple records

Why do these anomalies exist?

Because we've combined two themes (entity types) into one relation. This results in duplication, and an unnecessary dependency between the entities

Functional Dependencies and Keys

- Functional Dependency: The value of one attribute (the *determinant*) determines the value of another attribute

	SSN	Name	City	State	ZIP
t ₁	72163	John Smith	Chicago	IL	90101
t ₂	87991	Mark Green	LA	CA	90065
t ₃	87891	Mark Green	Los Angeles	CA	90065
t ₄	23212	Mary Clarke	LA	CA	90101

Functional Dependencies:

SSN → Name, City, State, ZIP

Name → SSN, City, State, ZIP

ZIP → State, City

Normalization

First Normal Form

- No multivalued attributes
- Every attribute value is atomic
- Derived Attributes should be removed
- Fig. 5-2a *is not* in 1st Normal Form (multivalued attributes) it is not a relation
- Fig. 5-2b *is* in 1st Normal form
- ***All relations are in 1st Normal Form***

	PROJ NUM	PROJ NAME	EMP NUM	EMP NAME	JOB CLASS	CHG HOUR	HOURS
▶	15	Evergreen	103	June E. Arbough	Elect. Engineer	\$84.50	23.8
			101	John G. News	Database Designer	\$105.00	19.4
			105	Alice K. Johnson *	Database Designer	\$105.00	35.7
			106	William Smithfield	Programmer	\$35.75	12.6
			102	David H. Senior	Systems Analyst	\$96.75	23.8
	18	Amber Wave	114	Annelise Jones	Applications Designer	\$48.10	24.6
			118	James J. Frommer	General Support	\$18.36	45.3
			104	Anne K. Ramoras *	Systems Analyst	\$96.75	32.4
			112	Darlene M. Smithson	DSS Analyst	\$45.95	44.0
	22	Rolling Tide	105	Alice K. Johnson	Database Designer	\$105.00	64.7
			104	Anne K. Ramoras	Systems Analyst	\$96.75	48.4
			113	Delbert K. Joenbrood *	Applications Designer	\$48.10	23.6
			111	Geoff B. Wabash	Clerical Support	\$26.87	22.0
			106	William Smithfield	Programmer	\$35.75	12.8
	25	Starflight	107	Maria D. Alonzo	Programmer	\$35.75	24.6
			115	Travis B. Bawangi	Systems Analyst	\$96.75	45.8
			101	John G. News *	Database Designer	\$105.00	56.3
			114	Annelise Jones	Applications Designer	\$48.10	33.1
			108	Ralph B. Washington	Systems Analyst	\$96.75	23.6
			118	James J. Frommer	General Support	\$18.36	30.5
			112	Darlene M. Smithson	DSS Analyst	\$45.95	41.4

FIGURE 4.1 A TABLE WHOSE STRUCTURE MATCHES THE REPORT FORMAT

	PROJ_NUM	PROJ_NAME	EMP_NUM	EMP_NAME	JOB_CLASS	CHG_HOUR	HOURS
▶	5	Evergreen	103	June E. Arbough	Elect. Engineer	\$84.50	23.8
			101	John G. News	Database Designer	\$105.00	19.4
			105	Alice K. Johnson *	Database Designer	\$105.00	35.7
			106	William Smithfield	Programmer	\$35.75	12.6
			102	David H. Senior	Systems Analyst	\$96.75	23.8

FIGURE 4.2 THE EVERGREEN DATA

Repeating group (any project can have a group of data entries) which should **not** to be appeared in relational table

Data Organization: 1NF

	PK		PK				
	PROJ_NUM	PROJ_NAME	EMP_NUM	EMP_NAME	JOB_CLASS	CHG_HOUR	HOURS
▶	15	Evergreen	103	June E. Arbough	Elect. Engineer	\$84.50	23.8
	15	Evergreen	101	John G. News	Database Designer	\$105.00	19.4
	15	Evergreen	105	Alice K. Johnson *	Database Designer	\$105.00	35.7
	15	Evergreen	106	William Smithfield	Programmer	\$35.75	12.5
	15	Evergreen	102	David H. Senior	Systems Analyst	\$96.75	23.9
	18	Amber Wave	114	Annelise Jones	Applications Designer	\$48.10	24.6
	18	Amber Wave	118	James J. Frommer	General Support	\$18.36	45.3
	18	Amber Wave	104	Anne K. Ramoras *	Systems Analyst	\$96.75	32.1
	18	Amber Wave	112	Darlene M. Smithson	DSS Analyst	\$45.95	44.0
	22	Rolling Tide	105	Alice K. Johnson	Database Designer	\$105.00	64.7
	22	Rolling Tide	104	Anne K. Ramoras	Systems Analyst	\$96.75	48.9
	22	Rolling Tide	113	Delbert K. Joenbrood *	Applications Designer	\$48.10	23.6
	22	Rolling Tide	111	Geoff B. Wabash	Clerical Support	\$26.87	22.5
	22	Rolling Tide	106	William Smithfield	Programmer	\$35.75	12.1
	25	Starflight	107	Maria D. Alonzo	Programmer	\$35.75	24.7
	25	Starflight	115	Travis B. Bawangi	Systems Analyst	\$96.75	45.8
	25	Starflight	101	John G. News *	Database Designer	\$105.00	56.3
	25	Starflight	114	Annelise Jones	Applications Designer	\$48.10	33.1
	25	Starflight	108	Ralph B. Washington	Systems Analyst	\$96.75	23.9
	25	Starflight	118	James J. Frommer	General Support	\$18.36	30.2
	25	Starflight	112	Darlene M. Smithson	DSS Analyst	\$45.95	41.4

FIGURE 4.3 DATA ORGANIZATION: FIRST NORMAL FORM

Conversion to 1NF

- Derived attributes must be removed
- Repeating groups must be eliminated
 - Proper primary key developed
 - Uniquely identifies attribute values (rows)
 - Combination of PROJ_NUM and EMP_NUM

Conversion to 1NF

- Repeating groups must be eliminated
 - **Dependencies** can be identified
 - A particular relationship between two attributes. For a given relation, attribute B is **functionally dependent** on attribute A if, for every valid value of A, that value of A uniquely determines the value of B.
 - A functional dependency exists when the value of one thing is fully determined by another. For example, given the relation EMP(empNo, empName, sal), attribute empName is functionally dependant on attribute empNo. If we know empNo, we also know the empName.

1NF Summarized

- All key attributes defined
- No repeating groups in table
- All attributes dependent on primary key

Second Normal Form

- 1NF *plus* every non-key attribute is fully functionally dependent on the ENTIRE primary key
 - Every non-key attribute must be defined by the entire key, not by only part of the key
 - No partial functional dependencies
- Fig. 5-2b is NOT in 2nd Normal Form (see fig 5-23b)

Fig 5.23(b) – Functional Dependencies in EMPLOYEE2

Dependency on entire primary key



Dependency on only *part* of the key

EmpID, CourseTitle → DateCompleted

EmpID → Name, DeptName, Salary

Therefore, NOT in 2nd Normal Form!!

Database Tables and Normalization

- Normalization stages
 - 1NF - First normal form
 - 2NF - Second normal form
 - 3NF - Third normal form
 - 4NF - Fourth normal form



Business
Bioinformatics
Statistical data

Better in
dependency

Worse in
performance
(I/O)

Desirable dependencies based on primary key
Less desirable dependencies

Partial

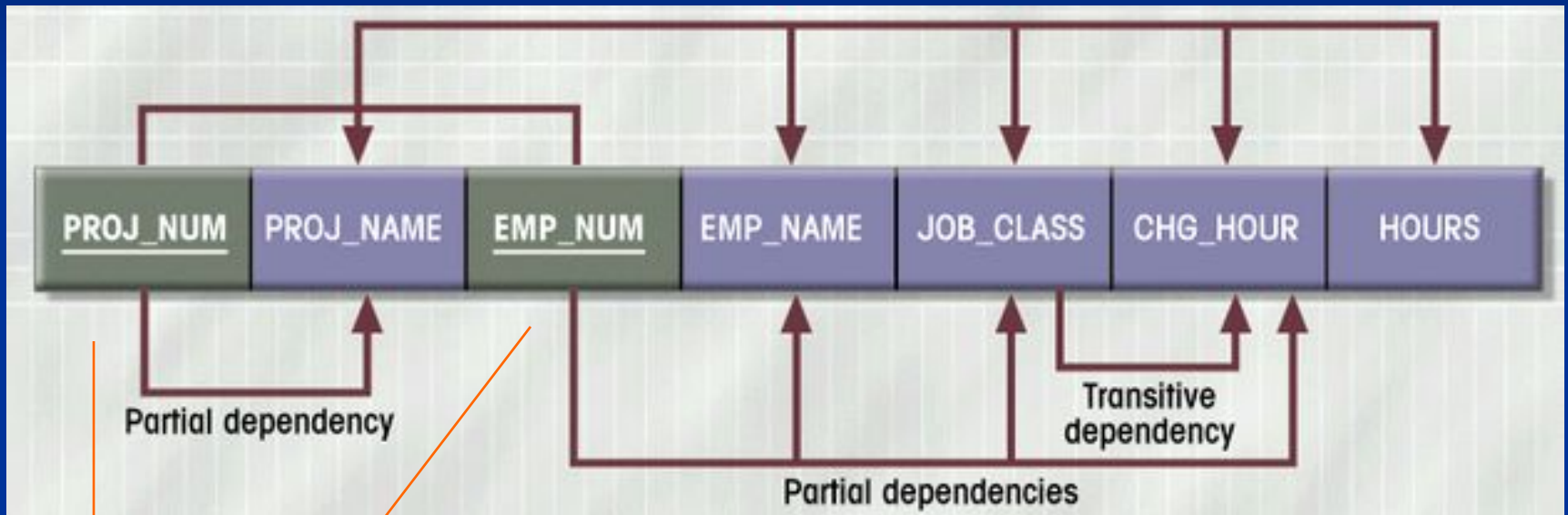
based on part of composite primary key

Transitive

one nonprime attribute depends on
another nonprime attribute

Dependency Diagram

Above: Desired Dependencies



Composite primary key

Below: Less Desired Dependencies

Figure 4.4

PROJ_NUM, EMP_NUM PROJ_NAME, EMP_NAME,
JOB_CLASS, CHG_HOUR, HOURS

↑
← DESIRED DEPENDENCIES

PROJ_NUM PROJ_NAME

← PARTIAL DEPENDENCIES

EMP_NUM EMP_NAME, JOB_CLASS, CHG_HOUR

JOB_CLASS -> CHG_HOUR

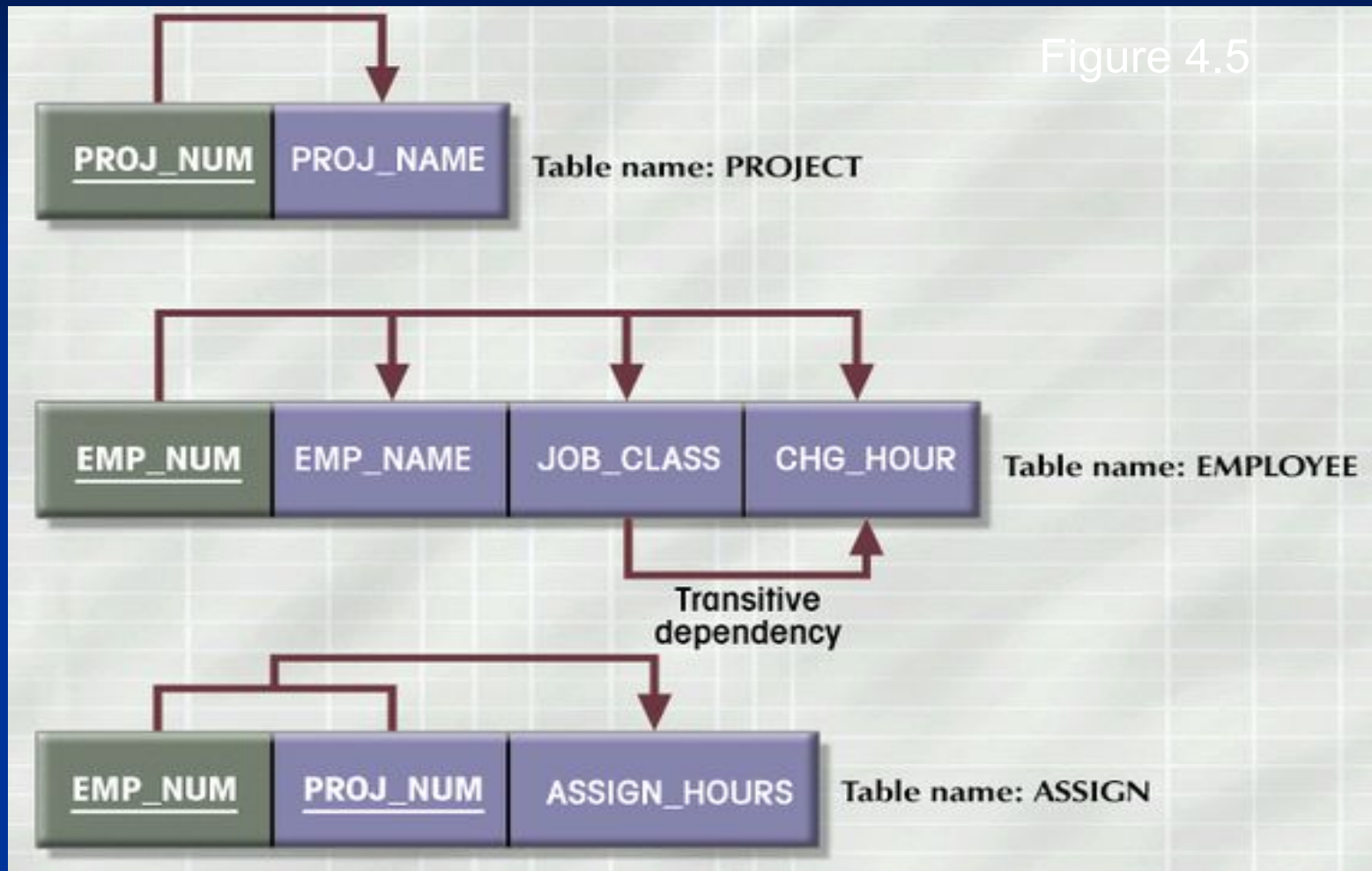
← TRANSITIVE DEPENDENCIES

Conversion to 2NF

- Start with 1NF format:
- Write each key component on separate line
- Write original key on last line
- Each component is new table
- Write dependent attributes after each key

```
PROJECT (PROJ_NUM, PROJ_NAME)
EMPLOYEE (EMP_NUM, EMP_NAME, JOB_CLASS, CHG_HOUR)
ASSIGN (PROJ_NUM, EMP_NUM, HOURS)
```

2NF Conversion Results



2NF Summarized

- In 1NF
- Includes no partial dependencies
 - No attribute dependent on a portion of primary key
- Still possible to exhibit transitive dependency
 - Attributes may be functionally dependent on nonkey attributes

Conversion to 3NF

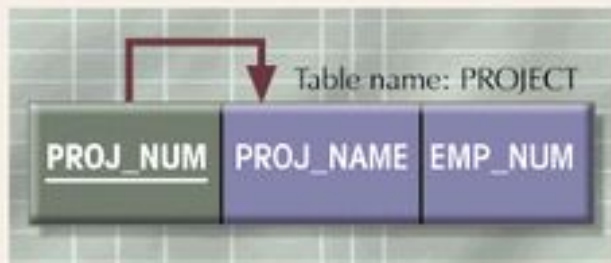
- Create separate table(s) to eliminate transitive functional dependencies

```
PROJECT (PROJ_NUM, PROJ_NAME)
ASSIGN (PROJ_NUM, EMP_NUM, HOURS)
EMPLOYEE (EMP_NUM, EMP_NAME, JOB_CLASS)
JOB (JOB_CLASS, CHG_HOUR)
```

3NF Summarized

- In 2NF
- Contains no transitive dependencies

Additional DB Enhancements



		PROJ_NUM	PROJ_NAME	EMP_NUM
▶	+	15	Evergreen	105
	+	18	Amber Wave	104
	+	22	Rolling Tide	113
	+	25	Starflight	101



		JOB_CODE	JOB_DESCRIPTION	JOB_CHG_HOUR
▶	+	500	Programmer	\$35.75
	+	501	Systems Analyst	\$96.75
	+	502	Database Designer	\$105.00
	+	503	Electrical Engineer	\$84.50
	+	504	Mechanical Engineer	\$67.90
	+	505	Civil Engineer	\$55.78
	+	506	Clerical Support	\$26.87
	+	507	DSS Analyst	\$45.95
	+	508	Applications Designer	\$48.10
	+	509	Bio Technician	\$34.55
	+	510	General Support	\$18.36

Figure 4.6

Table name: ASSIGN

ASSIGN_NUM	ASSIGN_DATE	PROJ_NUM	EMP_NUM	ASSIGN_HOURS	ASSIGN_CHG_HOUR	ASSIGN_CHARGE
------------	-------------	----------	---------	--------------	-----------------	---------------

ASSIGN_NUM	ASSIGN_DATE	PROJ_NUM	EMP_NUM	ASSIGN_HOURS	ASSIGN_CHG_HOUR	ASSIGN_CHARGE
1001	04-Mar-02	15	103	2.5	\$24.50	\$219.70
1002	04-Mar-02	18	118	1.4	\$18.36	\$25.70
1003	05-Mar-02	15	101	3.6	\$105.00	\$378.00
1004	05-Mar-02	22	113	2.5	\$48.10	\$120.25
1005	05-Mar-02	15	103	1.9	\$24.50	\$160.55
1006	05-Mar-02	25	115	4.2	\$96.75	\$406.35
1007	05-Mar-02	22	105	5.2	\$105.00	\$546.00
1008	05-Mar-02	25	101	1.7	\$105.00	\$178.50
1009	05-Mar-02	15	105	2.0	\$105.00	\$210.00
1010	06-Mar-02	15	102	3.8	\$96.75	\$367.65
1011	06-Mar-02	22	104	2.6	\$96.75	\$251.55
1012	06-Mar-02	15	101	2.3	\$105.00	\$241.50
1013	06-Mar-02	25	114	1.8	\$48.10	\$86.58
1014	06-Mar-02	22	111	4.0	\$26.87	\$107.48
1015	06-Mar-02	25	114	3.4	\$48.10	\$163.54
1016	06-Mar-02	18	112	1.2	\$45.95	\$55.14
1017	06-Mar-02	18	118	2.0	\$18.36	\$36.72
1018	06-Mar-02	18	104	2.6	\$96.75	\$251.55
1019	06-Mar-02	15	103	3.0	\$24.50	\$253.50
1020	07-Mar-02	22	105	2.7	\$105.00	\$283.50
1021	08-Mar-02	25	100	4.2	\$96.75	\$406.35
1022	07-Mar-02	25	114	5.8	\$48.10	\$278.98
1023	07-Mar-02	22	106	2.4	\$35.75	\$85.80

Table name: EMPLOYEE

EMP_NUM	EMP_LNAME	EMP_FNAME	EMP_INITIAL	EMP_HIREDATE	JOB_CODE
---------	-----------	-----------	-------------	--------------	----------

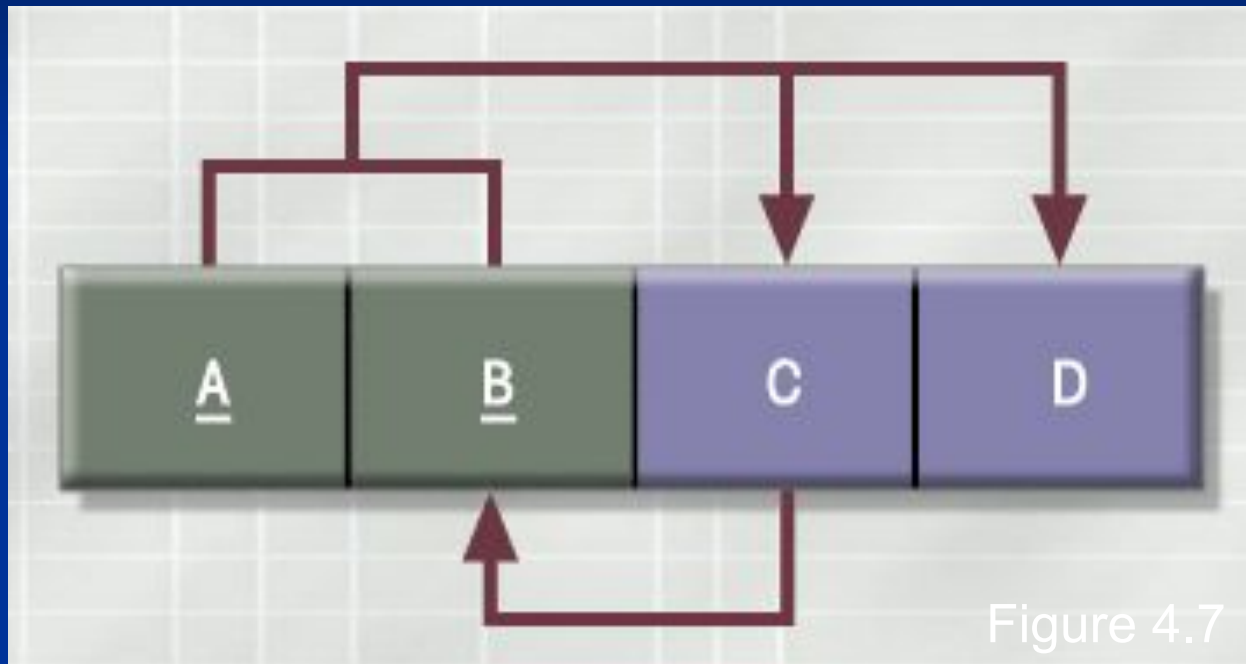
EMP_NUM	EMP_LNAME	EMP_FNAME	EMP_INITIAL	EMP_HIREDATE	JOB_CODE
101	Neeraj	John	O	09-Nov-00	502
102	Senar	David	H	12-Jul-99	501
103	Arbrough	June	E	01-Dec-94	503
104	Ramona	Anne	K	15-Nov-95	501
105	Johnston	Adce	K	01-Feb-91	502
106	Smithfield	William		22-Jun-01	500
107	Alonzo	Maria	O	10-Oct-91	500
108	Washington	Regis	B	22-Aug-99	501
109	Smith	Larry	W	18-Jul-95	501
110	Olenko	Gerald	A	11-Dec-93	505
111	Vilabach	Geoff	B	04-Apr-99	506
112	Smithson	Darlene	M	23-Oct-92	507
113	Joanbrood	Delbert	K	15-Nov-94	508
114	Jones	Annalise		20-Aug-91	508
115	Bewangi	Travis	B	25-Jan-90	501
116	Fratt	Gerald	L	05-Mar-95	510
117	Williamson	Angie	H	18-Jun-94	509
118	Frommer	James	J	04-Jan-02	510

FIGURE 4.6 THE COMPLETED DATABASE (CONTINUED)

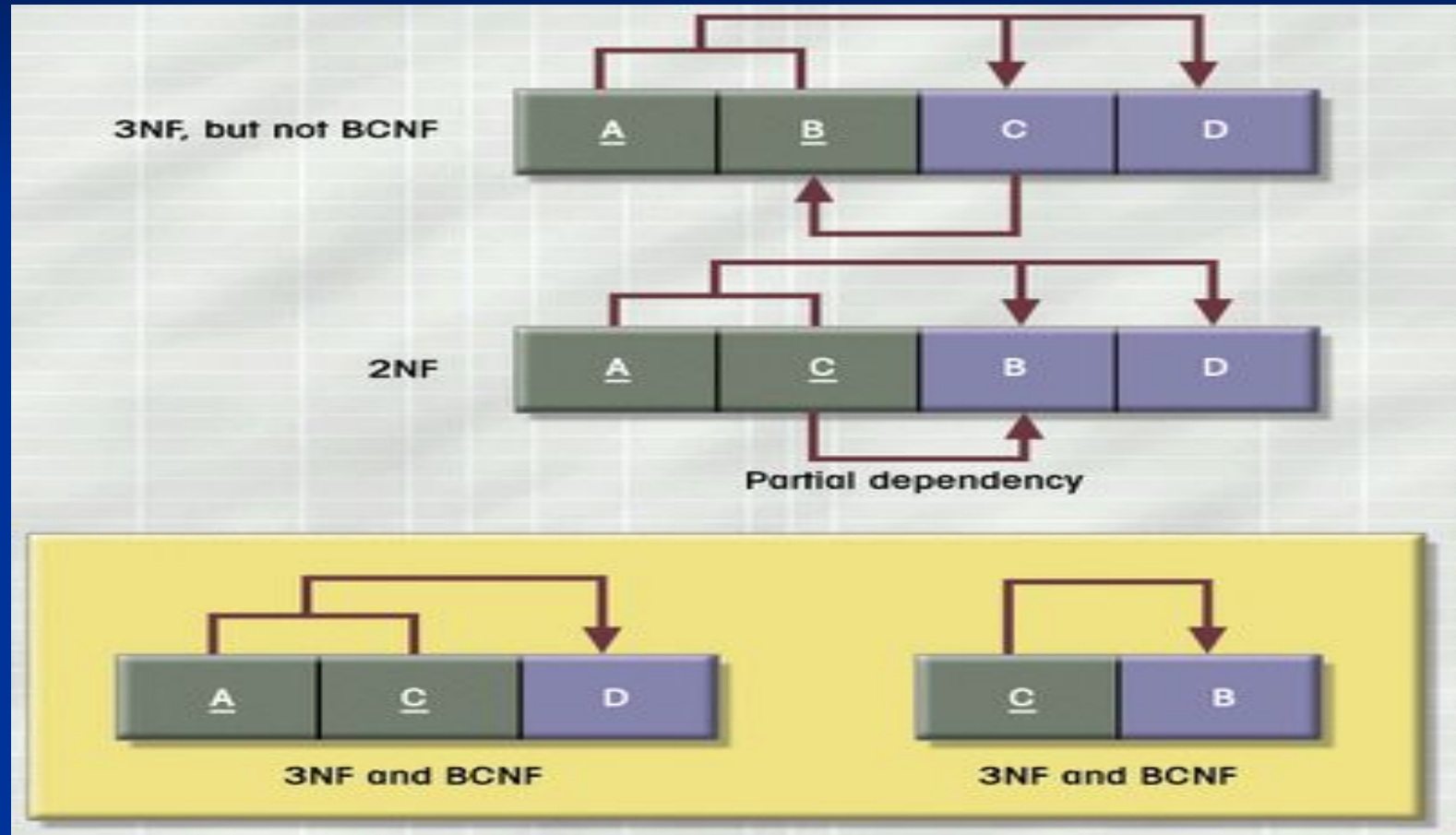
Boyce-Codd Normal Form (BCNF)

- Every determinant in the table is a candidate key
 - Determinant is attribute whose value determines other values in row
 - 3NF table with one candidate key is already in BCNF

3NF Table Not in BCNF



Decomposition of Table Structure to Meet BCNF



Example: BCNF conversion

TABLE 4.2 SAMPLE DATA FOR A BCNF CONVERSION

STU_ID	STAFF_ID	CLASS_CODE	ENROLL_GRADE
125	25	21334	A
125	20	32456	C
135	20	28458	B
144	25	27563	C
144	20	32456	B

Decomposition into BCNF

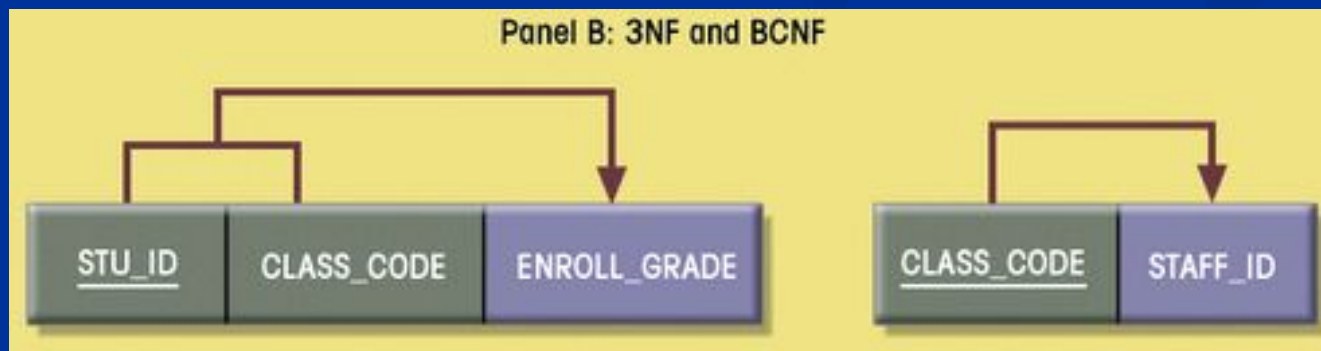
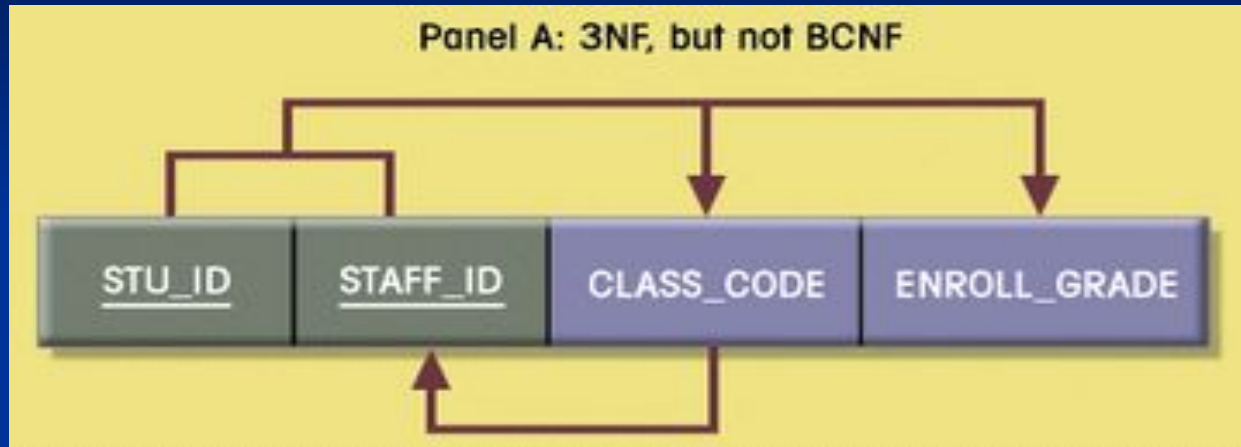


Figure 4.9

Normalization and Database Design

- Normalization should be part of the design process
- Make sure the proposed entities meet the required normal form before the table structures are created
- Used to redesign or modify the existing table structures.
- E-R Diagram provides macro view

Normalization and Database Design

- Normalization provides micro view of entities
 - Focuses on characteristics of specific entities
 - May yield additional entities
- Difficult to separate normalization from E-R diagramming
- Business rules must be determined

Normalization and Database Design

- Contracting company's example:

```
PROJECT (PROJ_NUM, PROJ_NAME)  
EMPLOYEE(EMP_NUM, EMP_LNAME, EMP_FNAME, EMP_INITIAL,  
          JOB_DESCRIPTION, JOB_CHG_HOUR);
```

Initial ERD for Contracting Company

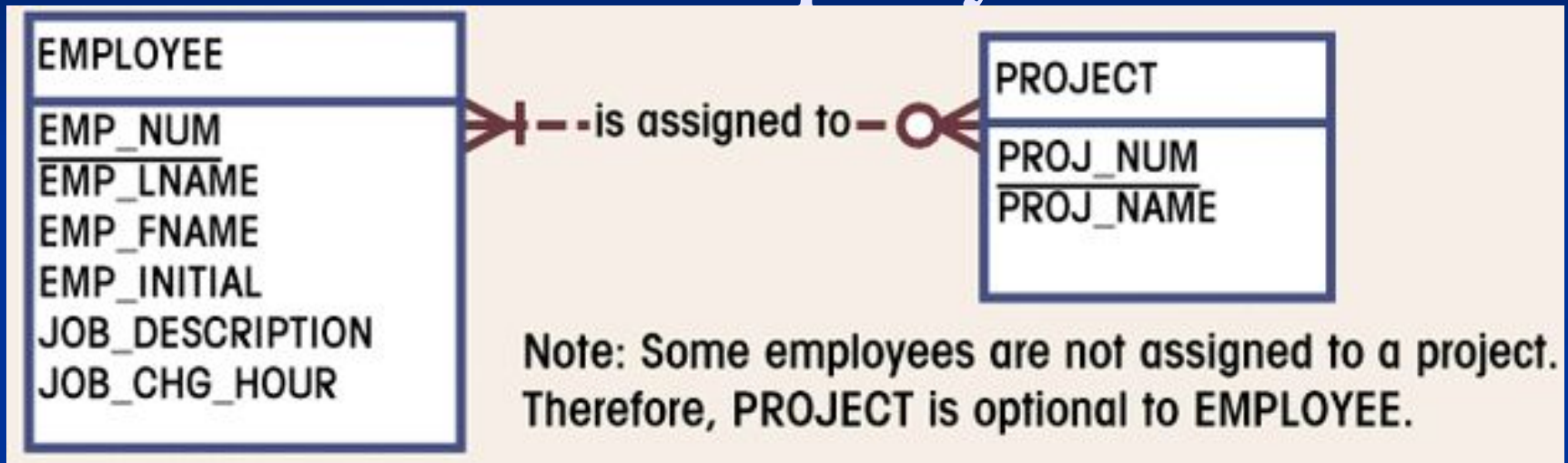


Figure 4.10

There is a transitive dependency

Already 3NF

Removal

```
PROJECT (PROJ_NUM, PROJ_NAME)
EMPLOYEE(EMP_NUM, EMP_LNAME,EMP_FNAME,EMP_INITAL,
        JOB_CODE)

JOB (JOB_CODE, JOB_DESCRIPTION, JOB_CHG_HOUR);
```

Modified ERD for Contracting Company

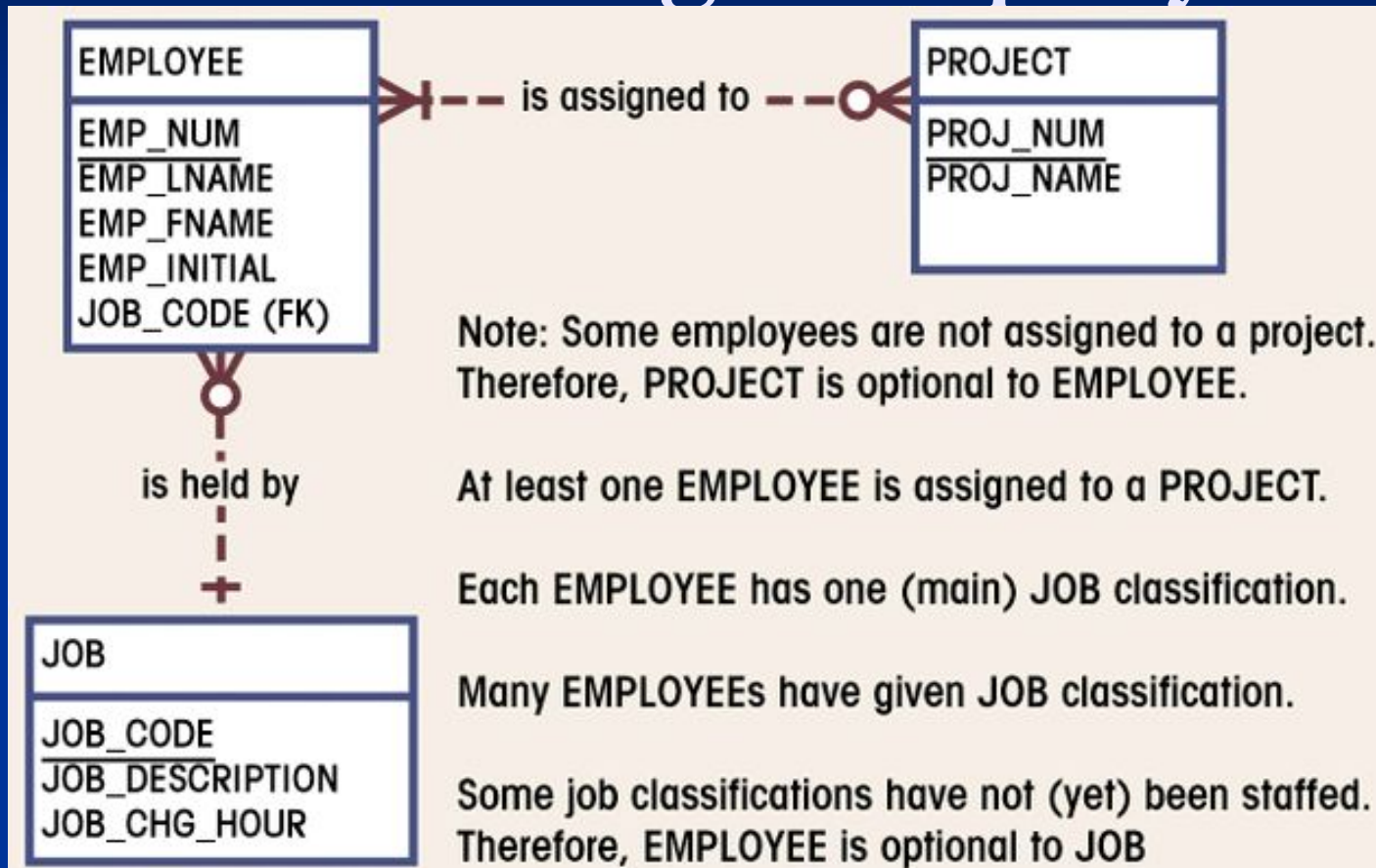


Figure 4.11

Final ERD for Contracting Company

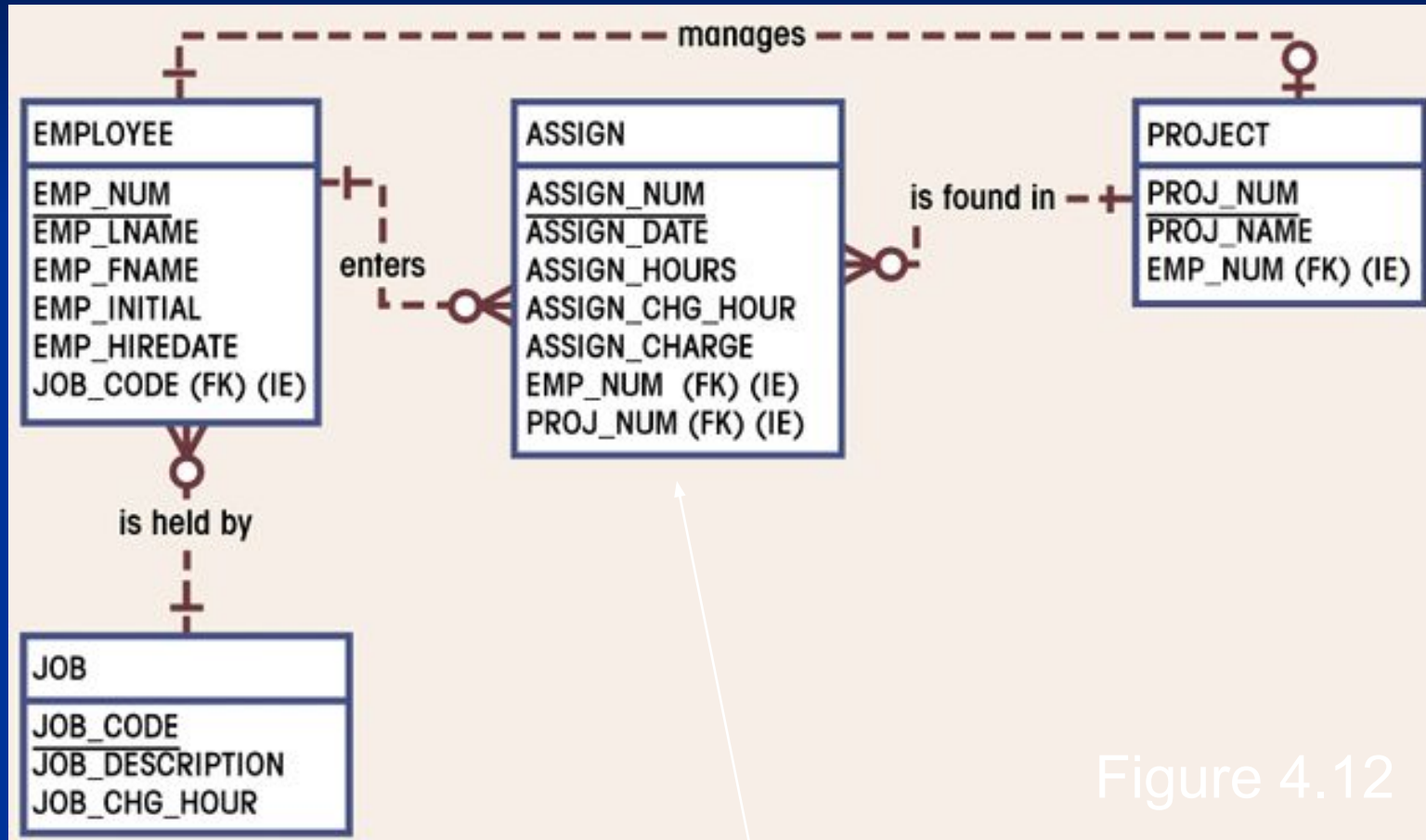


Figure 4.12

(M:N) converting to (1:M)

PROJECT (PROJ_NUM, PROJ_NAME, EMP_NUM)

EMPLOYEE(EMP_NUM, EMP_LNAME, EMP_FNAME, EMP_INITIAL,
EMP_HIREDATE, JOB_CODE)

JOB (JOB_CODE,, JOB_DESCRIPTION, JOB_CHG_HOUR);

ASSIGN((ASSIGN_NUM, ASSIGN_DATE, ASSIGN_HOURS,
ASSIGN_CHG_HOURS, ASSIGN_CHARGE, EMP_NUM, PROJ_NUM)

Table name: EMPLOYEE

EMP_NUM	EMP_LNAME	EMP_FNAME	EMP_INITIAL	EMP_HIREDATE	JOB_CODE
101	News	John	O	08-Nov-98	502
102	Senior	David	H	12-Jul-97	501
103	Arbough	June	E	01-Dec-94	503
104	Ramoses	Anne	K		
105	Johnson	Alice	K		
106	Smithfield	William			
107	Alonso	Maria	D		
108	Washington	Ralph	B		
109	Smith	Larry	W		
110	Olenko	Gerald	A		
111	Wabash	Geoff	B		
112	Smithson	Darlene	M		
113	Joelbrood	Delbert	K		
114	Jones	Annette			
115	Bawangi	Travis	B		
116	Pratt	Gerald	L	05-Mar-95	510
117	Williamson	Angie	H	19-Jun-94	509
118	Frommer	James	J	04-Jan-02	510

Database name: CONSTRUCTION_CO

Table name: JOB

JOB_CODE	JOB_DESCRIPTION	JOB_CHG_HOUR
500	Programmer	\$35.75
501	Systems Analyst	\$96.75
502	Database Designer	\$105.00
503	Electrical Engineer	\$84.50
504	Mechanical Engineer	\$67.90
505	Civil Engineer	\$55.78
506	Clerical Support	\$26.87
507	DSS Analyst	\$45.95
508	Applications Designer	\$48.10
509	Bio Technician	\$34.55
510	General Support	\$18.36

Table name: PROJECT

PROJ_NUM	PROJ_NAME	EMP_NUM
2	Evergreen	105
18	Amber Wave	104
22	Rolling Tide	113
25	Starlight	101

Table name: ASSIGN

ASSIGN_NUM	ASSIGN_DATE	PROJ_NUM	EMP_NUM	ASSIGN_HOURS	ASSIGN_CHG_HOUR	ASSIGN_CHARGE
1001	04-Mar-02	15	103	2.6	\$84.50	\$219.70
1002	04-Mar-02	18	118	1.4	\$18.36	\$25.70
1003	05-Mar-02	15	101	3.6	\$105.00	\$378.00
1004	05-Mar-02	22	113	2.5	\$48.10	\$120.25
1005	05-Mar-02	15	103	1.9	\$84.50	\$160.55
1006	05-Mar-02	25	115	4.2	\$96.75	\$406.35
1007	05-Mar-02	22	105	5.2	\$105.00	\$546.00
1008	05-Mar-02	25	101	1.7	\$105.00	\$178.50
1009	05-Mar-02	15	105	2.0	\$105.00	\$210.00
1010	06-Mar-02	15	102	3.8	\$96.75	\$367.65
1011	06-Mar-02	22	104	2.6	\$96.75	\$251.55
1012	06-Mar-02	15	101	2.3	\$105.00	\$241.50
1013	06-Mar-02	25	114	1.8	\$48.10	\$86.58
1014	06-Mar-02	22	111	4.0	\$26.87	\$107.48
1015	06-Mar-02	25	114	3.4	\$48.10	\$163.54
1016	06-Mar-02	18	112	1.2	\$45.95	\$55.14
1017	06-Mar-02	18	118	2.0	\$18.36	\$36.72
1018	06-Mar-02	18	104	2.6	\$96.75	\$251.55
1019	06-Mar-02	15	103	3.0	\$84.50	\$253.50
1020	07-Mar-02	22	105	2.7	\$105.00	\$283.50
1021	06-Mar-02	25	108	4.2	\$96.75	\$406.35
1022	07-Mar-02	25	114	5.8	\$48.10	\$278.98
1023	07-Mar-02	22	106	2.4	\$35.75	\$85.80

FIGURE 4.13 THE IMPLEMENTED DATABASE FOR THE CONTRACTING COMPANY

Denormalization

- Normalization is one of many database design goals
- Normalized table requirements
 - Additional processing
 - Loss of system speed

Denormalization

- Normalization purity is difficult to sustain due to conflict in:
 - Design efficiency
 - Information requirements
 - Processing

<u>RowID</u>	Course ID	Degree Program	Course Name, Credit	Semester
001	20101	BS	DB, 4	5th
002	20102,	BS	OOP, 3	2nd
003	20103	BS	DS, 3	3rd
004	30120	BS	FCS, 3	1st
005	20101	MIT	DB, 3	5 th
005	20101	MIT	DB, 3	5 th

<u>RowID</u>	Course ID	Degree Program	Course Name	Credit	Semester
001	20101	BS	DB	4	5 th
001	20102	BS	OOP	3	2 nd
003	20103	BS	DS	3	3 rd
004	30120	BS	FCS	3	1 st
005	20101	MIT	DB	4	5 th

<u>PID</u>	<u>DID</u>	<u>VID</u>	DName	DEd	PName	Tests	Diagnosis	Treatment
111	101	78	Asif	BDS	Abid	CBC	Dengue	Null
11	121	80	Ali	BDS	Kamil	LP	Null	Null
111	101	79	Asif	BDS	Abid	CBC	Dengue	Null
121	101	80	Asif	BDS	Zahid	LP	Null	Null
111	102	78	Amir	FRCS	Abid	CBC	Dengue	Null

[Company Name]

[Company Logo]

[Street Address]

[City, ST ZIP]

Phone: [000-000-0000]

Fax: [000-000-0000]

INVOICE

DATE 5/15/2012

INVOICE # 122

Customer ID XYZCompany

BILL TO:

Contact: John Doe

XYZ Company LLC

123 Somestreet NE

Anytown, CA 11111

Phone: (000) 000-0000

Fax: (000) 000-0000

SHIP TO (if different):

Mary Jane

XYZ Supply Company

123 Somestreet NE

Anytown, CA 11111

Due Date

6/14/2012

SALESPERSON	P.O. #	SHIP DATE	SHIP VIA	F.O.B.	TERMS
					Net 30

ITEM#	DESCRIPTION	QTY	UNIT PRICE	TOTAL
PCI 221	XYZ Base Product	15	1,234.00	18,510.00
PCI 221 abc	options: ABC	1	123.00	123.00
PCI 221 def	options: DEF	1	87.00	87.00
PCI 221 gh	options: GH	1	467.00	467.00
			-	-
			-	-
			-	-
			-	-
			-	-

Other Comments or Special Instructions

Please include the invoice number on your check

SUBTOTAL \$ 13,187.00

TAX RATE 6.875%

TAX \$ 1,313.11

S & H \$ -

OTHER \$ -

TOTAL \$ 20,506.11

If you have any questions about this invoice, please contact
 [Contact Name] Phone: [000-000-0000], [name@xyzcompany.com]

Make all checks payable to
 [Company Name]

Thank You For Your Business!

Please detach the portion below and return it with your payment.

REMITTANCE

[Company Name]

[Street Address]

[City, ST ZIP]

Phone: [000-000-0000]

Fax: [000-000-0000]

DATE 5/15/2012

INVOICE # 122

Customer ID XYZCompany

Due Date 6/14/2012

AMOUNT ENCLOSED

1 Tax invoice

2 Windows to Fit Pty Ltd
ABN: 32 123 456 789 3

15 Burshag Road
Festler NSW 2755

4 Date: 1 August 2013

To: Building Company 8
254 Burshag Road
Festler NSW 2755

Qty	Description of supply	Unit price	GST	Total
5 50	Window frames	\$150	\$15	\$8,250
10	Deadlocks	\$40	\$4	\$440
			6	
TOTAL AMOUNT PAYABLE				\$8,690

The total price includes GST

7